

Rakesh Ranjan

Graduate Student Member, IEEE

Next Generation Wireless Communication Lab

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Education

- 2019 – 2024 **Doctor of Philosophy (Ph.D) in Communication and Signal Processing**
Electronics Engineering Department
Indian Institute of Technology (IIT), Dhanbad, India, 826004.
Current Status: Thesis Submitted on 1st October 2024.
Research Area: *Beyond 5G/6G Wireless Communication Technology such as Intelligent Reflecting Surface (IRS)-aided Communication Systems, Integrated Sensing and Communication (ISAC).*
Thesis Title: *Efficient Phase Shift Optimization Framework for Intelligent Reflecting Surface (IRS)-Aided Wireless Systems.*
Supervisor: Dr. Himanshu Bhusan Mishra, Associate Professor.
Co-supervisor: Dr. Samrat Mukhopadhyay, Assistant Professor.
- 2017 – 2019 **Master of Technology (M.Tech.) in Communication and Signal Processing.**
Electronics Engineering Department
Indian Institute of Technology (IIT), Dhanbad, India, 826004.
Thesis Title: *Adaptive Wideband Beamforming With Sensors Delay Lines.*
- 2009 – 2013 **Bachelor of Technology in Electronics and Communication Engineering.**
The University of Burdwan - Burdwan, West Bengal, India, 713104.

Awards and Achievements

- 2021 **Two-year Senior Research Fellowship (SRF)** at IIT Dhanbad from Ministry of Education, Govt. of India.
- 2019 **Two-year Junior Research fellowship (JRF)** at IIT Dhanbad from Ministry of Education, Govt. of India.
- Institute fellowship grant from MHRD, Govt. of India.**
- 2018 **Graduate Aptitude Test in Engineering (GATE)** 2018 qualified.
- 2017 **Graduate Aptitude Test in Engineering (GATE)** 2017 qualified.
- 2016 **Graduate Aptitude Test in Engineering (GATE)** 2016 qualified.
- 2015 **Graduate Aptitude Test in Engineering (GATE)** 2015 qualified.
- 2013 Received **Bachelor of Engineering Degree with First Class Honours.**

Research Interests

- 2019 – 2024 Intelligent Reflecting Surface (IRS)-aided wireless communication systems, Physical layer security in IRS-aided systems, IRS-Integrated Sensing and Communication (ISAC) Systems.

Skills

Languages  Hindi (Professional), English (Professional).
Softwares  Matlab/SIMULINK[®], L^AT_EX, Microsoft Office.

References

- 1 **Dr. Himanshu Bhusan Mishra**
Associate Professor, Department of Electronics Engineering
Indian Institute of Technology (IIT), Dhanbad, Jharkhand-826004, India.
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- 2 **Dr. Samrat Mukhopadhyay**
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- 3 **Dr. Suraj Suman**
Assistant Professor, Department of Electrical Engineering
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Research Publications

Journals/ Conferences

1. **Rakesh. Ranjan**, A. Bhattacharya, H.B. Mishra and S. Mukhopadhyay, "A Gradient Ascent Based Low Complexity Rate Maximization Algorithm for Intelligent Reflecting Surface-Aided OFDM Systems" *IEEE Communications Letters*, vol. 27, no. 8, pp. 2083-2087, Aug. 2023, doi: 10.1109/LCOMM.2023.3289865.
2. **Rakesh. Ranjan**, A. Bhattacharya, H. B. Mishra and S. Mukhopadhyay, "A Low-Complexity Phase Shift Optimization to Achieve Security in IRS-Assisted MISO Systems," *IEEE National Conference on Communications (NCC) 2024, IIT-Madras, Chennai, India*, pp. 1-6, doi: 10.1109/NCC60321.2024.10485931.
3. **Rakesh. Ranjan**, A. Bhattacharya, H.B. Mishra and S. Mukhopadhyay, "Achievable Rate Maximization for Reconfigurable Intelligent Surface-Assisted MIMO Systems: A Novel Low-Complexity Solution," *Elsevier Signal Processing* (under revision).
4. **Rakesh. Ranjan**, A. Bhattacharya, M.V. Katwe, H.B. Mishra, K.Singh and D. W. Kwan Ng, "A Robust STAR-RIS-aided MU-MIMO System for Integrated Sensing and Communication," *IEEE Transactions on Wireless Communications* (under revision).